

Is the Finite-Shadow Language a Faithful Compression of Collatz Dynamics?

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Abstract

A referee asks: *how do we know the finite-shadow language captures all arithmetic mechanisms available to Collatz, rather than only the mechanisms visible to the surviving candidates inside the framework?* We answer in four parts. (1) The language is *lossless*, not a projection: the valuation-word coding is a homeomorphic conjugacy of the 2-adic Syracuse system onto the full shift, so no mechanism is discarded at the symbolic level; what the language cannot express is exactly one thing — membership in $\mathbb{N} \subset \mathbb{Z}_2$ — and the program imports that through an explicit, enumerable list of archimedean mechanisms (clocks, counting, the drift identity, Baker). (2) Within the finite-modulus category the language is provably *complete*: every observable of a bounded orbit window modulo any Q coprime to 6 factors exactly through (one residue, the letters), and conditional on any word cylinder the residues at all odd places are exactly equidistributed — only the 2-adic coordinate feeds the word; every other finite place is a passenger. (3) The empirical hidden-channel test is null: multiplicative invariants (ω , squarefreeness, divisibility, primality) of deep $\{1, 2\}$ -survivors match size-matched controls exactly. (4) The “reversed-flow” concern conflates presentation with content: the load-bearing theorems of the program are $\forall$ -integer statements in Tao’s direction; and the audit of arXiv:2603.11066 shows three independently built frameworks translating term-for-term into one another and hitting the *same* residual wall — the language-invariance one expects of a faithful compression, not the framework-relativity one would fear.

1 The question, made precise

Call an *arithmetic mechanism* any constraint usable in a proof: an observable of orbits together with a theorem about it. The framework under audit speaks the language of valuation words $b_n = v_2(3x_n + 1)$, finite character shadows, 2-adic alignments (clocks), and band counting. The referee’s worry: perhaps Collatz dynamics supports mechanisms invisible in this language, so that narrowing the surviving profile *inside* the framework overstates progress. Faithfulness splits into three claims, each addressable: (a) the coding loses nothing; (b) within its declared categories the language is complete; (c) what lies outside is isolated explicitly, not silently compressed away. All verification counts refer to `search/faithfulness.py`.

2 Losslessness: conjugacy, not projection

Theorem 1 (Lossless coding). *The word map $x \mapsto (b_n(x))_{n \geq 0}$ is a homeomorphic conjugacy from the accelerated Syracuse system on the odd 2-adics onto the full shift on $\{1, 2, 3, \dots\}^{\mathbb{N}}$, carrying Haar measure to the Bernoulli measure $\mathbb{P}[b = k] = 2^{-k}$. A length- k prefix w pins exactly the residue class $x \bmod 2^{B_w+1}$ and nothing more. (Verified: all 4845 realizable length-4 words each occupy exactly one class, total mass $0.9987 \rightarrow 1$; letter marginals are exactly 2^{-k} by residue counting.)*

This is the Bernstein–Lagarias conjugacy in our coordinates; our cylinder lemma is its quantitative form. Consequence: *no* mechanism is lost in passing to the symbolic language — the 2-adic system and the word system are the same object. The language has exactly one blind spot, and it is not a mechanism but a *subset*: $\mathbb{N} \subset \mathbb{Z}_2$ (dense, measure zero, non-invariant). The framework does not pretend to express it symbolically; it imports it through four explicit interfaces, each an exact theorem: the ghost clocks (positivity caps $a \leq \log_2(|2^K - 3^\ell|x + C|)$), without-replacement counting (an acyclic orbit visits each integer once), the drift identity (exact, with error $\sum \log_2(1 + \frac{1}{3x_n})$ explicitly bounded), and the Baker/ S -unit inputs. Any mechanism the language misses must live in this declared residual — which is precisely the part the program treats as open.

3 Completeness within the finite-modulus category

Theorem 2 (k -point shadow completeness). *For any Q coprime to 6 and any k : along any orbit, $x_{n+i} \bmod Q = (3^i x_n + A_i(w)) 2^{-B_i(w)} \bmod Q$ for $i \leq k$, so every Q -periodic observable of a k -step window factors exactly through $(x_n \bmod Q, b_n, \dots, b_{n+k-1})$; and for every multiplicative character, $\chi(x_{n+1})\chi(2)^{b_n} = \chi(3x_n + 1)$ exactly — the keystone identity. (Verified: 2000 random windows, $Q \in \{5, 7, 11, 13, 25, 35\}$.)*

Theorem 3 (CRT orthogonality). *Conditional on any word cylinder (an arithmetic progression modulo 2^{B+1}), the residues modulo any odd m are exactly equidistributed. Hence the word reads only the 2-adic coordinate of x ; every other finite place evolves as an explicit skew product driven by the word and carries no information about future letters beyond it. (Verified: 40 random depth-18 cylinders, exact uniformity mod 3, 5, 7.)*

Together: *within the category of finite-modulus observables of bounded windows — the natural home of “congruence mechanisms” — the shadow language is complete, provably and exactly.* A congruence mechanism unavailable in the language does not exist.

4 The hidden-channel test

Mechanisms outside the congruence category would have to use the multiplicative or global structure of orbit values (factorization, primality, smoothness). Theorem 3 does not force these to be neutral, so we test: among the 2800 odd $n_0 < 2^{22}$ surviving 23 letters in $\{1, 2\}$, against size-matched random odd controls:

	mean $\omega(n)$	$\mathbb{P}[\text{squarefree}]$	$\mathbb{P}[3 \mid n]$	$\mathbb{P}[\text{prime}]$
survivors	2.480	0.806	0.334	0.144
controls	2.478	0.815	0.319	0.144

(the squarefree density of *odd* integers is $8/\pi^2 = 0.811$; both populations sit on it). Null within sampling error: no multiplicative channel into survival is detected. A deviation here would have been a genuine mechanism outside the language; finding none is the falsifiable content of faithfulness, and the test is cheap to repeat at larger depth.

We record one instructive artifact: ensemble letter autocorrelation measured naively on 2^{42} -sized seeds gave $+0.021$, an apparent violation of the Bernoulli prediction — caused entirely by absorption into the trivial cycle (the constant $b = 2$ tail). Measured on pre-absorption windows it is -0.003 ± 0.003 . The framework predicted both numbers; apparent unfaithfulness was, on inspection, the $+1$ ghost.

5 The direction of flow

The concern that the program only narrows a surviving profile — reversing Tao’s all-integers flow — describes its *presentation*, not its theorems. The load-bearing results are universal statements over all integers or all acyclic orbits, in exactly Tao’s direction:

$\forall$ -statement	form
narrow-band dichotomy	all orbits, $\text{dwell} \leq CY^{\theta(D)}$
logarithmic dwell at ratio ≤ 2	all acyclic orbits
escape rate $O(X \log X)$	all divergent orbits
upcrossing bound 2^{k-3}	all divergent orbits
two-clock identities, parse	all odd integers, exact
universal ghost clock, supply	all signatures, all orbits
exact counts $N_D(p)$, $(3/4)^D$ law	all words/classes
entropy deficit $\theta_{12} < 0.9791$	all bands, all widths
Haar i.i.d. round laws	all residue classes

The “surviving profile” is the *complement* of these universal constraints, stated for orientation. Compressing the complement is how universal theorems report progress.

6 Language-invariance of the residual

Finally, the strongest available evidence that the compression is canonical rather than self-serving: independent frameworks built in different languages translate into ours term by term. In the audit of arXiv:2603.11066: their burst-gap calculus *is* the two-clock parse (their $k = v_2(n+1)$ is our depth g ; their Persistent Exit Lemma is our parity handoff at $h = 3$, verified 4000/4000); their mean-gap threshold 1.71 is our $1/\log_2(3/2)$; their two open hypotheses are our two calibration channels; their CIC is the parity- $\equiv 1$ extreme of our calibration conjecture; their phantom-cycle census is our ghost catalogue with the same roots $C_\sigma/(2^K - 3^\ell)$. Tao’s framework (Syracuse random variables, 3-adic drivers, logarithmic density) likewise meets the same final gap — almost-all versus all, distributional versus pointwise, annealed versus quenched. Three languages, one wall: the obstruction is a property of the dynamics, not of any framework. Were the finite-shadow language an unfaithful narrowing, its wall would be an artifact located elsewhere than the others’; instead all three walls coincide under explicit dictionaries.

We also record, for honesty, where the framework’s *stochastic heuristics* (as opposed to its language) are known to deviate from exact dynamics: the defect-cocycle C-suite measured exact-vs-annealed gaps (operator gaps, excursion-length-dependent shadow rates). None of the unconditional theorems above rely on annealed models; the measured gaps are themselves expressed — and were found — *inside* the language.

Verdict

Faithful compression = lossless coding + provable completeness on the declared categories + explicitly isolated residual. All three hold: the conjugacy (Theorem 1), the finite-modulus completeness theorems (Theorems 2 and 3, with null hidden-channel tests), and the residual — $\mathbb{N}$ -positivity, the ghost gap — which is not compressed away but stated as the open core, where every known framework independently arrives. What would falsify faithfulness is specific and testable: a statistic outside the

language that predicts survival beyond its CRT projection. We looked; there is none so far, and the test suite ships with the program.

References

- [1] J. C. Lagarias, *The $3x + 1$ problem and its generalizations*, and D. Bernstein, *A non-iterative 2-adic statement of the $3N + 1$ conjecture* (conjugacy).
- [2] T. Tao, *Almost all orbits of the Collatz map attain almost bounded values*, Forum Math. Pi (2022).
- [3] E. Y. Chang, arXiv:2603.11066v6 (2026), audited in the ghost-clock companion note.
- [4] J. A. Janik, *Finite Spectral Shadows for the Collatz Valuation Cocycle* (2026), and companion notes (two clocks; universal ghost clock; narrow-band; CIC round chain).